

Closed Native Verification Systems (CNVS): Complete Formal Theory and Emergent Security Theorem

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Abstract

This paper introduces Closed Native Verification Systems (CNVS), a formal class of verification systems in which state membership is verification-dependent rather than primitive. The theory combines a typed structural universe, a type-preserving recursive decomposition, a random assignment of non-uniform fragments, limited verifier knowledge, and a global invariant check, first applying data convergence in a local verification environment. A conditional probabilistic theorem limits unauthorized reconstruction with a binomial tail and produces exponential suppression when the reconstruction threshold exceeds the expected number of useful compromised fragments. An asymptotic scaling theorem shows that security can increase as the system expands in the presence of explicit assumptions. The technical difficulty of reconstruction outside the system arises from the strictly local knowledge available to each verifier.

1. Primitive Domains and Typed Universe

Let $B = \{0,1\}$ and B^ be the set of all finite binary strings. Define the primitive domains:*

$Id \subseteq B^$ (identifiers),*

$D \subseteq B^$ (declared data),*

$At \subseteq B^$ (attribute classes),*

$Val \subseteq B^$ (externally reported values),*

P (finite set of logical properties),

τ (finite set of structural types).

The structural universe is:

$U = Id \times D \times At \times Val \times \text{Pow}(P)$.

Each element $s \in U$ has the form $s = (id, d, a, v, p)$.

2. State Space

A system state at time t is:

$S(t) = (E(t), R(t), C, V_L)$,

where $E(t) \subseteq U$ is the set of admitted elements, $R(t) \subseteq E(t) \times E(t)$ is a directed relation set, $C = \{c_1, \dots, c_m\}$ is a finite family of invariant predicates, and V_L is the local verification function.

3. Local Verification

The local verification function is:

$$V_L : Id \times D \times At \times Val \rightarrow \{0,1\} \times Pow(P).$$

For $x = (id, d, a, v)$, let:

$$V_L(x) = (b, p),$$

where

$$b = Form(id, d, a, v) \wedge Conv_a(d, v).$$

Local admissibility is:

$$Adm_L(s, t) = 1 \Leftrightarrow \pi 1(V_L(s)) = 1.$$

4. Recursive Decomposition and Reconstruction

The decomposition operator is:

$$D_t : U \rightarrow Pow_fin(U).$$

For every admissible element s :

$$D_t(s) = \{F1, \dots, Fk\}, 1 \leq k < \infty.$$

Type preservation:

$$\forall s' \in D_t(s), type(s') = type(s).$$

Reconstruction:

$$Rec_t(D_t(s)) = s$$

up to canonical encoding.

5. Non-Uniform Fragmentation

There exist fragments Fa and Fb such that:

$$|I(Fa)| \neq |I(Fb)|.$$

Hence decomposition is intentionally non-uniform. Some fragments contain more information than others.

6. Knowledge Restriction

For each external verifier v_j assigned to attribute a_j :

$$I_{\min}(a_j) \leq K_V(j,t) \ll I_N(i,t) \ll I_G(t)$$

where $I_{\min}(a_j)$ is the minimum information required to preserve the semantic meaning of the assigned verification task.

Knowledge cannot converge to zero absolutely, since complete loss of information would eliminate the verifier's ability to perform the assigned task.

7. Information Density

For a fragment F_j :

$$\rho_N(j,i,t) = |I_F(j,t)| / |I_N(i,t)|,$$

$$\rho_G(j,t) = |I_F(j,t)| / |I_G(t)|,$$

$$\rho_C(j,i,t) = \rho_N(j,i,t)^\alpha \rho_G(j,t)^\beta,$$

where $\alpha, \beta \in [0,1]$ and $\alpha + \beta = 1$.

8. Random Assignment

Terminal fragments are assigned to external verifiers by:

$$A_t : T(t) \times \mathcal{E} \rightarrow V_{\text{ext}}(t),$$

where $\mathcal{E}$ is a probability space and $T(t)$ is the set of terminal fragments.

9. Axiom I: Verification-Dependent Existence

For every element s and time t :

$$s \in E(t) \Rightarrow \text{Adm}_L(s,t) = 1.$$

Existence inside the system is a derived consequence of successful verification.

10. Axiom II: Randomized Non-Uniform Fragmentation

Every admissible element is recursively decomposed into non-uniform fragments that are assigned randomly:

$$\begin{aligned} D_t(s) &= \{F_1, \dots, F_k\}, \\ \exists Fa, Fb : |I(Fa)| &\neq |I(Fb)|, \\ A_t : T(t) \times \mathcal{E} &\rightarrow V_{ext}(t), \\ Rec_t(D_t(s)) &= s. \end{aligned}$$

11. Axiom III: Non-Reducibility of Global Validity

Global validity is not reducible to local validity:

$$V_G(S) \neq \bigwedge_{s \in E} V_L(s).$$

Instead:

$$V_G(S) = LocalOK(E) \wedge Cons_R(R, E) \wedge Inv_C(S).$$

11a. State Transition Law

A candidate successor state is admitted only if it satisfies global verification:

$S(t+1)$ exists in the system if and only if $V_G(S(t+1)) = 1$.

Equivalently, any candidate state S' such that $V_G(S') = 0$ is rejected and cannot become a valid system state.

Thus, state evolution is verification-dependent and constrained by all structural invariants.

12. Fundamental Lemmas

Lemma 1. Local admissibility is necessary for state membership.

Lemma 2. Local validity is insufficient for global validity.

Lemma 3. Type is preserved across decomposition depth.

Lemma 4. Reconstruction conserves payload information.

Lemma 5. Metadata overhead permits encoded expansion.

Lemma 6. Average fragment density decreases under controlled subdivision.

13. Principle of Knowledge Restriction

For every external verifier:

$$I_{\min}(a_j) \leq K_V(j,t) \ll I_G(t).$$

If K_V remains bounded while $|I_G|$ grows, then:

$$|K_V(j,t)| / |I_G(t)| \rightarrow 0.$$

Thus each verifier retains the minimum semantic information required to perform the assigned task while controlling an asymptotically vanishing fraction of the total global information.

Equivalently, the verifier's knowledge provides negligible information about the native information content, so that

$$H(I_N | K_V(j,t)) \approx H(I_N),$$

or, equivalently,

$$I(I_N ; K_V(j,t)) \approx 0.$$

14. Probabilistic Security Model

Let:

k = number of terminal fragments,

r = reconstruction threshold,

C = colluding verifier count,

Q = total verifier count,

$$q = C/Q,$$

λ_j = probability that a compromised fragment is useful.

Define $X_j = 1$ if fragment j is compromised and useful, and

$$X = \sum_j X_j.$$

Unauthorized reconstruction occurs when:

$$\text{Rec}^* = \{X \geq r\}.$$

$$\omega(D_i) = I(G; F_i) / H(G)$$

$$\lambda_{\max} = \max_j \lambda_j$$

$$\lambda_j = \Pr(\omega(D_i) \geq \delta)$$

$$p_{\text{eff}} = q \lambda_{\max}$$

$\tau \in (0, 1)$,

where τ is the minimum normalized amount of useful information required for unauthorized reconstruction.

15. Conditional Security Theorem

If X is stochastically dominated by $\text{Bin}(k, p_{\text{eff}})$, then:

$$\Pr(\text{Rec}^*) \leq \sum_{i=r}^k \binom{k}{i} p_{\text{eff}}^i (1-p_{\text{eff}})^{k-i}.$$

Proof.

By definition, unauthorized reconstruction requires at least r useful compromised fragments.

Under stochastic domination, $X \leq_{\text{st}} Y$ where $Y \sim \text{Bin}(k, p_{\text{eff}})$. Therefore:

$$\Pr(X \geq r) \leq \Pr(Y \geq r),$$

which equals the stated binomial upper tail. QED.

16. Exponential Chernoff Bound

Let $\mu = k p_{\text{eff}}$. If $r > \mu$, then:

$$\Pr(\text{Rec}^*) \leq \exp(-(r-\mu)^2 / (2\mu + (r-\mu))).$$

Proof.

Apply the multiplicative Chernoff bound to $Y \sim \text{Bin}(k, p_{\text{eff}})$. QED.

17. Emergent Security Scaling Theorem

Let n denote system scale. Assume:

$$k(n) \rightarrow \infty,$$

$p_{\text{eff}}(n)$ remains bounded or decreases,

$$r(n) - k(n)p_{\text{eff}}(n) \rightarrow +\infty.$$

Then:

$$\Pr(\text{Rec}^*_n) \rightarrow 0.$$

Proof.

Substitute the scaling assumptions into the Chernoff exponent. The exponent diverges to $+\infty$, hence the probability tends to zero. QED.

In the weighted formulation, if

$q \bar{\omega} < \tau$,

where q is the probability that a fragment is assigned to a colluding verifier and $\bar{\omega}$ is the average informational weight of compromised fragments, then:

$$q \bar{\omega} < \tau \implies \lim_{n \rightarrow \infty} P [\sum_{i=1}^n Y_i \omega(D_i) \geq \tau n] = 0.$$

18. Interpretation

Under the stated assumptions, system growth increases fragmentation and epistemic restriction. As a result, unauthorized reconstruction becomes exponentially less probable.

Security is therefore an emergent property of scale.

In an idealized uniform model, the number of possible fragment assignments grows factorially with the number of terminal fragments. The corresponding combinatorial entropy is $H_n = \log_2(k(n)!)$.

Therefore, if $k(n) \rightarrow \infty$, then $H_n \rightarrow \infty$, indicating that the uncertainty faced by an external adversary increases as the system expands.

18a. Corollary: Global Veto Property

If at least one essential contribution to a global invariant remains unknown or uncontrolled by colluding verifiers, reconstruction of a globally valid state is not guaranteed.

Any incorrect estimation of that contribution may cause one or more global invariants to fail, yielding:

$$V_G(S) = 0$$

and rejection of the candidate state.

18b. Critical Fragment Principle

The Security of the System does not depend solely on the percentage of Verifiers under adversarial control, but on whether the adversary possesses all essential information required to reconstruct a globally valid state.

A Fragment is said to be essential if its absence prevents the construction of at least one

globally admissible configuration satisfying the System invariants.

A Fragment is said to be non-inferable if it cannot be derived from the information already available to the adversary.

Formally, let K_{adv} denote the set of Fragments known to the adversary. If there exists at least one Fragment F_h such that

$F_h \notin K_{adv}$

and F_h is both essential and non-inferable, then successful reconstruction of a globally valid state is not guaranteed.

Equivalently,

$\exists F_h \notin K_{adv}$ such that F_h is essential and non-inferable $\implies \Pr(Rec^) < 1$.*

This principle emphasizes that the decisive factor is not the absolute proportion of colluding Verifiers, but the existence of at least one indispensable Fragment that remains outside adversarial control.

Consequently, even extensive collusion does not imply certain success unless the adversary acquires every essential Fragment required for globally coherent reconstruction.

18c. Gold Allocation Example

Suppose 100 orders should sum to 100 grams of gold.

If colluding agents know 99 orders totaling 99 grams, the remaining unknown order may represent any admissible quantity.

An incorrect estimate may produce a total inconsistent with the invariant, for example 102 grams instead of 100 grams.

Therefore:

$V_G(S) = 0$.

As the value of $|I_G|$ increases, the difficulty of reconstructing the missing information while preserving global invariant consistency increases correspondingly.

19. Scientific Positioning

CNVS combines concepts from information theory, probability theory, cryptography, formal methods, graph theory, and epistemic logic into a unified mathematical framework.

20. Open Problems

1. *Formal leakage models.*
2. *Dynamic collusion models.*
3. *Cryptographic reductions.*
4. *Information-theoretic lower bounds.*
5. *Empirical validation.*

Appendix A. Minimal Formal Definition

A structure

$M = (U, E, R, C, D, A, V_L, V_G, Rec, K)$

is a Closed Native Verification System if and only if:

1. *U is a typed finite-binary universe.*
2. *Membership requires local admissibility.*
3. *Global validity requires relation consistency and invariants.*
4. *Decomposition is finite, recursive, and reconstructible.*
5. *Verifier knowledge is strictly restricted.*
6. *Terminal assignments are randomized.*
7. *Invalid elements and states are rejected.*

END DOCUMENT

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